



Cambridge (CIE) IGCSE Computer Science



Your notes

Robotics

Contents

* Robotics



Your notes

Robotics

Robotics

What is robotics?

- Robotics is the principle of a robot **carrying out a task** by **following a precise set of programmed instructions**
- Robots can be categorised into **two** groups:

Dumb robots	Smart robots
Repeat the same programmed instructions over and over again (no AI)	Carries out more complex tasks and can adapt and learn (AI)
E.g. Car assembly line	E.g. Assisting surgeons in delicate procedures

- The development of artificial intelligence, including the increased use of machine learning and robotics raises **ethical** and **legal** issues such as:
 - Accountability**
 - Safety**
 - Algorithmic bias**
 - Legal liability**

Characteristics of a Robot

What are the characteristics of a robot?

- Robots share **three** common characteristics:
 - Mechanical structure**
 - Electrical components**
 - Programmable**

Mechanical structure	Electrical components	Programmable
----------------------	-----------------------	--------------

A physical body that can move and interact with the environment	Sensors, microprocessors and actuators	Designed to follow a precise set of instructions
---	--	--

The Role of Robots

What is the role of robots?

- Robots play a crucial role in many areas such as:

Scenario	Role	Advantages	Disadvantages
Industry	Quality inspector	Advanced vision to inspect products for defaults, improving quality control	May not work on defects that might require human judgement
Transport	Warehouse worker	Efficiency in finding, moving and packing, less reliable on manual labour	High infrastructure costs and potential safety concerns for human workers sharing the same space
Agriculture	Planting & seeding machine	Precise planting (optimum depths and distances)	High upfront costs, may struggle with uneven terrain and changing weather conditions
Medicine	Rehabilitation therapist	Personalised physical therapy exercises to aid recovery	Doesn't provide emotional support
Domestic	Sous chef	Assists with meal prep by doing basic duties (chopping, measuring and following basic recipes)	Limited ability and capabilities
Entertainment	Educational robot	Interactive learning experiences for children, increased engagement	High cost, cannot replace the guidance of human teachers



Worked Example



Your notes

A company uses robots in its factory to manufacture restaurant tables and chairs

(i) One characteristic of a robot is that they contain electrical components.

State two other characteristics of a robot.

[2]

(ii) Give **two advantages** to company **employees** of using robots to manufacture tables and chairs.

[2]

(ii) Give **one disadvantage** to the company's owners of using robots to manufacture table and chairs

[1]

Answers

(i)

- Mechanical structure
- Programmable

(ii)

- Employees don't need to lift heavy items
- Employees can be protected from dangerous tasks
- Employees can utilise their skills in other tasks
- Employees don't need to perform repetitive/mundane tasks

(iii)

- Expensive to install/purchase/setup
- High ongoing costs/maintenance costs
- May deskill the workforce
- If they malfunction, production may stop